Spotting Sepsis in the Waiting Room: Explainable Risk Prediction from Triage Data

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58 vs **27** sepsis cases in 100 found by the model vs the SIRS rule, **flagging the same 5.3% of arrivals**

0.925 AUROC from triage data alone (95% CI 0.919–0.932)

56% of sepsis cases fall in the red tier, which flags **5% of arrivals**

Abstract

Can triage data alone flag sepsis? On 409,216 adult ED visits (MIMIC-IV-ED), XGBoost ranked risk in unseen patients with AUROC 0.925 and, at equal workload, found more cases than SIRS, qSOFA, NEWS or nurse acuity, with three reasons for every flag.

1. Background

Sepsis is linked to one in five deaths worldwide [1]. Patients who look stable at triage can deteriorate in a crowded waiting room. NSW Health's SAFE-WAIT trial (Western Sydney, from 2027) shows the problem matters here.

2. Objectives

- Rank risk in patients the model has never seen
- Beat SIRS, qSOFA, NEWS and nurse acuity at equal workload
- Give a calibrated low / moderate / high tier
- Show the reasons behind each flag
- Check performance across patient groups

3. Methodology

1. **Data: MIMIC-IV-ED v2.2**
   425,087 ED visits, Boston, 2011–2019 [2], [3]
2. **Cohort: 409,216 adult visits**
   196,624 patients; 10,092 (2.5%) with sepsis coded
3. **62 triage-time features**
   Vitals, age, sex, complaint, arrival, past visits. No blood tests
4. **Split by patient 70 / 15 / 15**
   Train · calibrate and set cut-offs · test once, then locked
5. **Three models**
   Logistic regression, random forest, XGBoost [4]; 5-fold CV grouped by patient
6. **Evaluate and explain**
   1,000 patient bootstrap resamples for 95% CIs; SHAP [5]; subgroup audit

Fig. 1. Pipeline. Only information known at triage enters the model.

4. System Architecture

Triage form
Vitals, complaint, arrival, history

→

Feature builder
Same code as training: no drift

↓

XGBoost
Isotonic-calibrated risk, tier cut-offs

←

Traffic light + reasons
Tier, risk and top three SHAP reasons

Fig. 2. Prototype: an offline web app that scores one patient with the trained model.

Key Technologies

Python · XGBoost · SHAP · scikit-learn · pandas · matplotlib · pytest · HTML / JS

5. Results

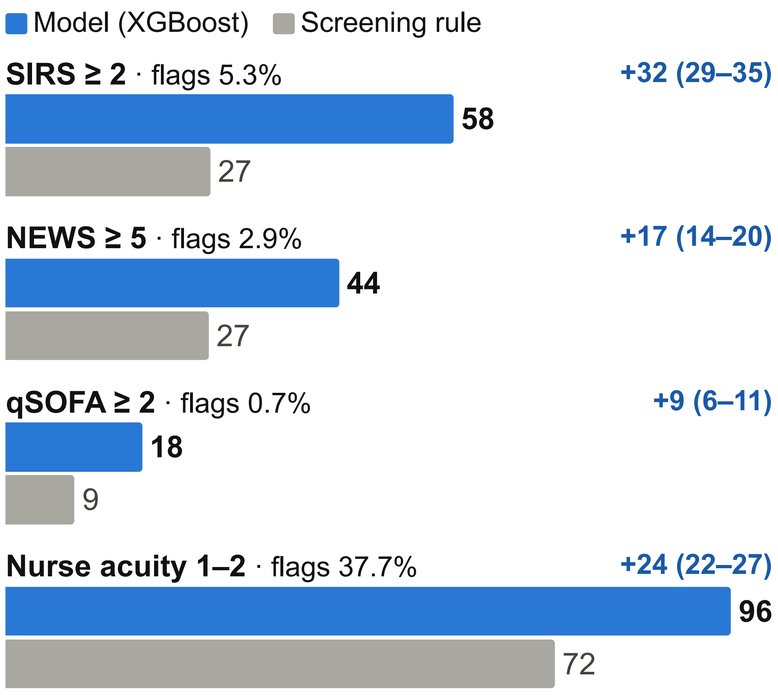


Fig. 3. At the same number of alerts, the model found more sepsis cases (per 100) than every rule. Difference in points, 95% CI. Test set: 61,690 visits, 1,508 cases.

TABLE I
Ranking performance on the test set (sepsis rate 2.4%)

Method	AUROC (95% CI)	AUPRC
XGBoost	**0.925** (0.919–0.932)	**0.360**
Logistic regression	0.919 (0.912–0.926)	0.338
Random forest	0.917 (0.910–0.924)	0.332
NEWS (triage)	0.807 (0.795–0.819)	0.148
Nurse acuity	0.710 (0.697–0.722)	0.057
SIRS (triage)	0.709 (0.695–0.721)	0.065
qSOFA (triage)	0.667 (0.655–0.680)	0.076

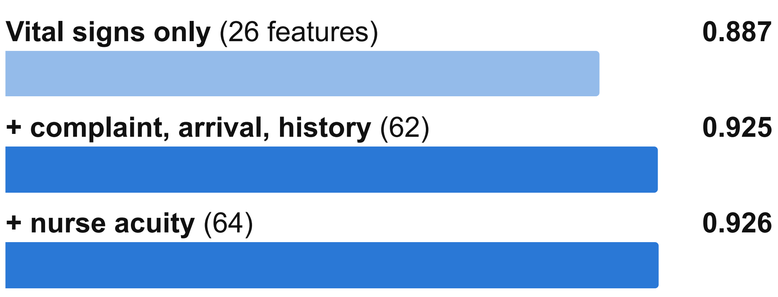


Fig. 4. Context beyond vital signs lifts AUROC by 0.039; the nurse's acuity score adds almost nothing (+0.001).

6. Traffic-Light Risk Tiers



Tier	Arrivals	Had sepsis	Predicted
High	4.9%	28 in 100	28 in 100
Moderate	15.0%	5.2 in 100	4.9 in 100
Low	80.2%	0.4 in 100	0.5 in 100

Fig. 5. The red tier is 5% of arrivals but holds 56% of sepsis cases, and predicted risk matches what happened (calibration slope 1.02).

⚠ **Research prototype, not for clinical use**

7. What Drives the Risk

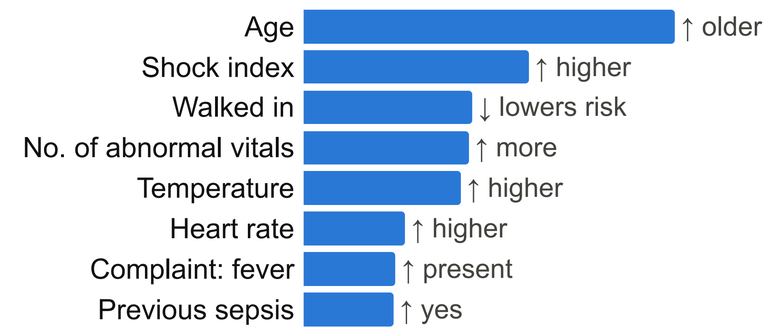


Fig. 6. Top drivers by mean |SHAP|. They are clinically recognisable, and the ranking agreed with permutation importance (ρ = 0.76).

8. Demonstration

⚠ High risk

Estimated sepsis risk 53%

Top reasons

1. **Complaint: fever** = 1 ↑ raises risk
2. **Shock index** = 1.08 ↑ raises risk
3. **Number of abnormal vitals** = 3 ↑ raises risk

Complaint read as: fever, cough.

In the held-out test set, this tier held 4.9% of arrivals and 28% of them had sepsis recorded.

Tiers: high = top 5% of arrivals by model score, moderate = next 15%. Research prototype, not for clinical use.

Fig. 7. Prototype output for an invented patient: 78-year-old by ambulance with fever and cough, temperature 38.6 °C, heart rate 112, blood pressure 104/62. Every flag shows its top three reasons.

9. Who It Works For

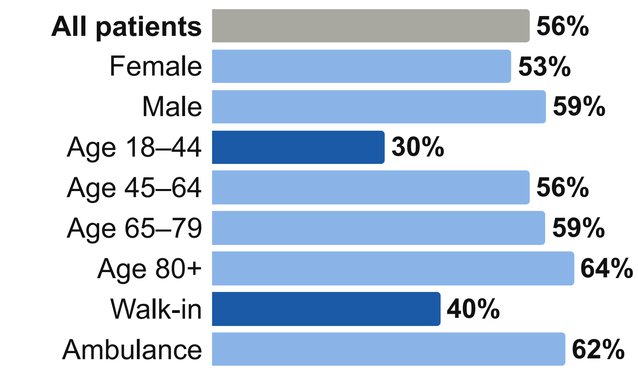


Fig. 8. Share of sepsis cases put in the red tier. Ranking held in all 20 groups (AUROC 0.89–0.94), but young adults and walk-ins, where sepsis is rarer, are flagged less often.

10. Conclusions

- Triage data alone found twice as many sepsis cases as SIRS at the same workload.
- Logistic regression came within 0.006 AUROC: a transparent model is nearly as good.
- **Limits:** one US hospital, retrospective; sepsis defined by diagnosis codes.
- **Next:** ethics approval, de-identified NSW data, and an equity audit including Aboriginal and Torres Strait Islander patients.

References

1. K. E. Rudd *et al.*, "Global, regional, and national sepsis incidence and mortality, 1990–2017," *Lancet*, vol. 395, no. 10219, pp. 200–211, 2020.
2. A. E. W. Johnson *et al.*, "MIMIC-IV, a freely accessible electronic health record dataset," *Sci. Data*, vol. 10, Art. no. 1, 2023.
3. A. Johnson *et al.*, "MIMIC-IV-ED (version 2.2)," PhysioNet, 2023, doi: 10.13026/5ntk-km72.
4. T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD*, 2016, pp. 785–794.
5. S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. NeurIPS*, 2017, pp. 4765–4774.

Acknowledgements

Data accessed through PhysioNet credentialed access. Patient-level data never left the analysis laptop; only aggregate results are shared, with counts below 10 suppressed.

Sepsis-3 code lists adapted from MIMIC-Sepsis (MIT licence). Parts of the code were written with help from Claude (Anthropic); the team made and checked every analysis choice and number.

GitHub repository
github.com/rupeshghimire7/Sepsis_Research_Poster